cations. They are working on conductive, 3D-printable gelatins that could be used to make circuits for sensors that could be swallowed.

Egg-based electronics

Egg white, also known as egg albumen, has very good dielectric properties along with high transparency and high elasticity. It is an ideal material for organic electronics as it is easy to process, readily accessible, and available at a lower cost.

Heat transforms albumen from a soluble liquid to an insoluble solid as its proteins undergo thermal irreversible denaturation, unfolding from linear chains to form networks.

These networks are held together by interactions between amino acids on different protein chains. For example, the thiol functional groups (containing sulpur and hydrogen) of cysteine amino acids in different protein chains form disulfide bonds.

The new research marks that egg albumen can be used to make resistive memories that operate based on changes in resistance rather than electric current as a next-generation alternative to the silicon-based memories that dominate today's electronics. These resistive memories will have potential advantages such as higher speeds, higher densities, and smaller sizes.

Albumen's dielectric constant is higher than that of most polymers used in OFETs, making it a potential insulating material. This is not the first time that egg albumen has been incorporated into electronic devices, but previously also the albumen from chicken and duck eggs has been used in transistors and other devices as the dielectric (insulating) layer.

In a recent study, researchers have found that on mixing egg albumen with hydrogen peroxide, a series of chemical reactions occur that transform the biomaterial into an active film useful to make transparent, flexible resistive memory devices. Scientists have already developed a simple and low-cost fabrication process to make an albumen dielectric layer for OFETs. First, by separating egg white and yolk using a mesh spoon. Then, by creating a thin albumen film by spin-coating it onto a pre-patterned indium tin oxide substrate.

To obtain a smooth and dense dielectric film, the substrate is heated gradually from 80 to 140 degrees Celsius in a vacuum chamber by adding a layer of organic materials and depositing a metal layer through shadow masks.

The dielectric film is one of the main components of resistive memories, and here egg albumen-based film is used, which is normally insulating but can be made conducting by applying voltage. Switching between the states of high and low electrical resistance corresponds to switching between the memory's 'off' and 'on' states, respectively.

The researchers have been able to demonstrate that the resistance of egg albumen material can be made switchable by mixing it with a ten per cent hydrogen peroxide solution. Egg albumen contains more than forty different proteins that are linked together by weak chemical bonds, and deep inside these proteins are large numbers of iron, sodium, and potassium ions.

These ions such as $Fe3+$, $Na+$, and $K+$ are always connected with protein chains in the chicken egg albumen, and so cannot work efficiently when charges are injected. But on treating with ten per cent hydrogen peroxide solution, ions are exposed outside of protein chains. Thus, the resistive switching memory properties of hydrogen peroxide-modified egg albumen film are efficiently improved compared to those of pristine egg albumen. The hydrogen peroxide easily breaks the bonds that are holding proteins together. This denatures proteins and critically exposes the ions.

These ions, which are positively charged, then act as traps that capture negatively charged electrons that are injected when a voltage is applied. When the trap levels are low (few or no electrons), the dielectric material behaves as an insulator, and the memory is in 'off' state. When a negative voltage is applied, it causes the traps to fill with electrons, the material becomes conducting, and the memory switches to 'on' state. In order to reset the memory, a positive voltage is applied, which releases the electrons from the traps and returns the memory to its 'off' state.

Research teams have experimented with different materials to produce dissolvable electronics, including DNA, other types of proteins, and metals, but now the egg-based memristor is developed by researchers. An electronic device made of egg protein albumin has been created by researchers, paving the way for dissolvable sensors that could be used inside the human body.

The device, known as the memristor or memory resistor, is used to regulate the flow of electric current and can also remember charges. It has magnesium and tungsten-based electrodes that are incorporated on a thin film of albumin, a protein found in egg whites.

When used in dry laboratory conditions, the device's performance remained consistent for more than three months, but in water, however, it dissolved almost completely within three days. The ability of this biomaterial to dissolve without trace shows promise for possible medical applications.

Summary

Although great progress and breakthroughs have been made regarding the new material's application and structure design, the mechanism of resistive switching memory is still not completely clear. So, further research has to be continued to investigate the mechanism of resistive switching memory and at the same time, flexible, wearable, and water-dissolution resistive switching memory cells need to be developed using organic-modified egg albumen.

In summary, albumen functions well as a low-cost dielectric material for OFETs. Chicken albumen as a biomaterial can be used as the gate dielectric in OFETs. The albumen does not need further extraction or purification and is easily obtained, processed, and less expensive than other biomaterials for OFET devices. The entire process for preparing albumen dielectric uses properties of natural albumen without synthesis.

Albumen's dielectric constant is higher than that of most polymers used in OFETs, making it a potential insulating material. Albumen dielectrics are also suitable for fabricating flexible OFETs, which are lighter and more durable. Such devices have many more potential applications than those with glass substrates, particularly for portable electronics. Research suggests that albumen dielectrics work well in flexible OFETs and complementary inverters. EFY